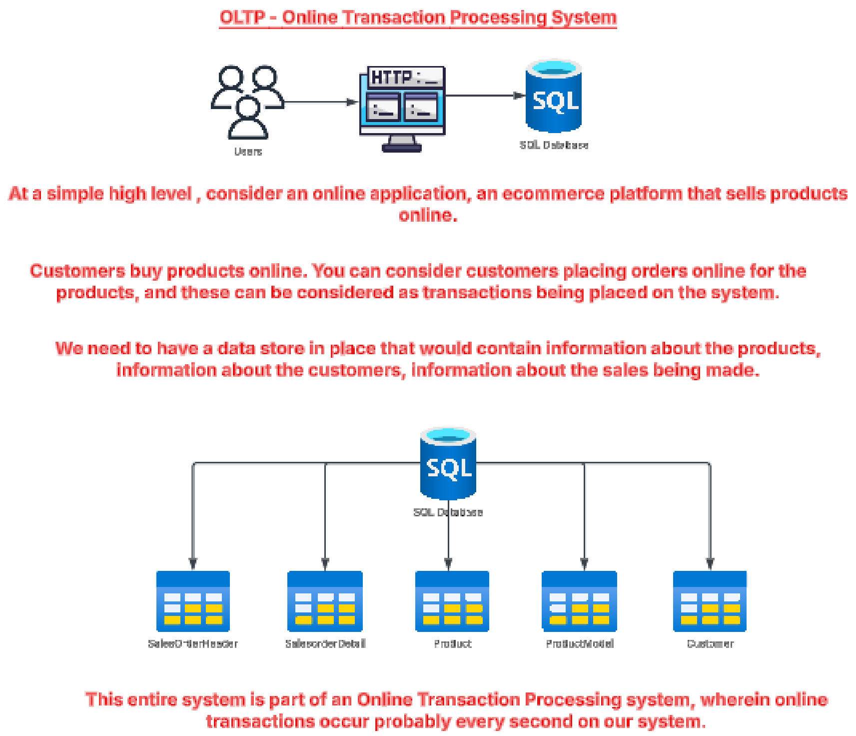
Implement and manage semantic models - Power BI Desktop

Online Analytical Processing Systems



OLAP - Online Analytical Processing System

We now want to have data store in place that we can use to perform analysis on our data.

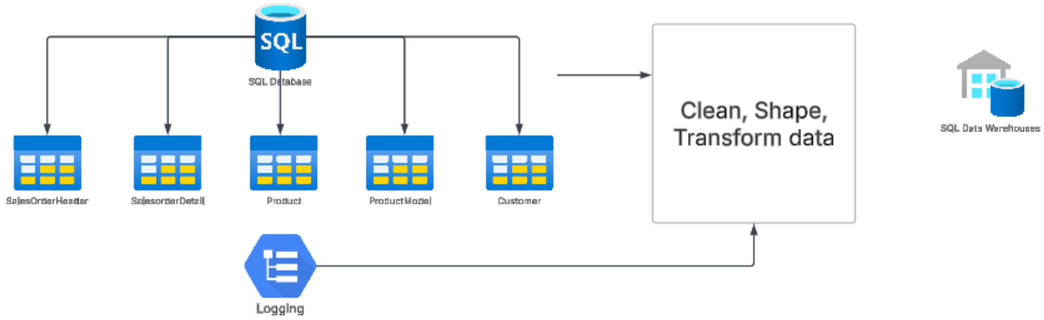
For example, performance of sales over time, which regions did the best etc.

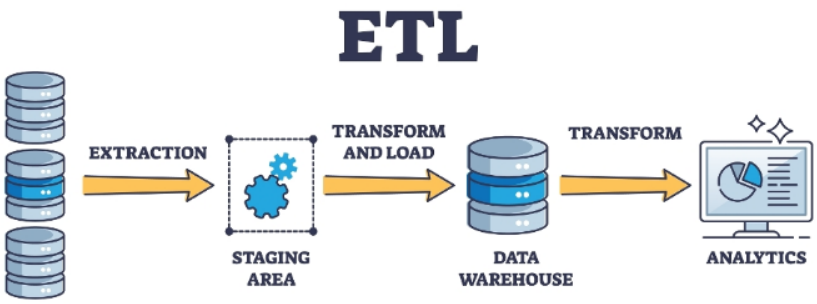
Here we need to run queries over our existing historical data.

We might also club the queries with data coming in from other data sources because we want to get a good view or different perspectives to our data.

We don't want to perform the queries on our existing OLTP data store, because we don't want to put any performance strain via the queries on our existing system. It might create a performance issue for our application.

So we need to create a seperate data store environment that we can use to perform analysis of our data.





You can take in data from a variety of data sources, shape the data in the way that you want. Transform the data and then load the data onto a data store system that supports a data warehouse.

You have services and tools that allow you to host and query a data warehouse. Here the underlying system is built in such a way that it allows you to host a large data set. And also allows efficient querying of data across the large data set.

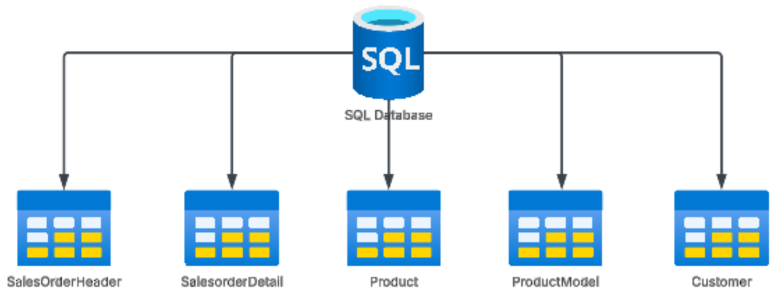
What goes into building a data warehouse

When we build a data warehouse, we need to model our data in such a way that it becomes efficient to actually gain insights from the data.

We can actually create a data model, which is just a way that we structure the data. We also organize our data and define relationships between the different data entities.

Dimension Data Modeling - This is a design approach used to build data warehouses.

Here you design your tables as Fact and Dimension tables.



When you build tables to store data for your OLTP system, you design the structure in such a way that it becomes efficient to modify data within and across tables.

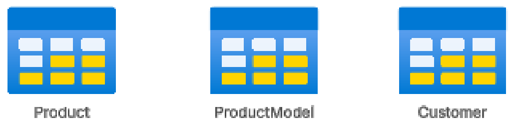
But with an OLAP system, we are more focused on querying for data rather than modifying data. Yes, we do need to keep on adding more data as it becomes available in the data warehouse. But that is more of appending data rather than updating existing data.

We are more focused on querying data to gain more insights.

A Fact table is meant to store quantitative data, that is data that can be measured.

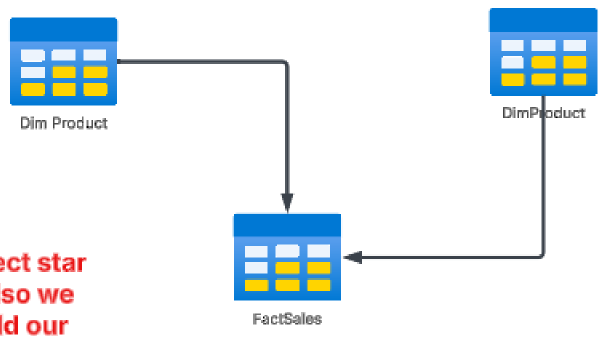
If we were to look at the data in our Sales based table, this would contribute to Fact-based data. Here each sale made is a fact, an event that has occured in the business.

Dimension tables are used to used to present some context to the facts.



We might want to gain some insights from a Product perspective or a Customer perspective to our fact-based data.

We then design something known as a star schema, which is model of the relationships between the fact and dimension table.



Not the perfect star as of yet. Also we have to build our dimension and fact tables based on our actual table data.

With Power BI Desktop we will start building queries based on sources of data available in comma seperated files.

We will work towards building a very simple data model based on the star schema which will contain fact and dimension tables.

In the end we want to build a data model or a semantic data model. This semantic data model would contain our data, relationships between our data tables, calculations , reports etc.

Power BI Desktop - Merge Queries

Power Query

We use Power Query in the Power Query Editor to connect and transform our source data.

This is a data transformation and data preparation engine.

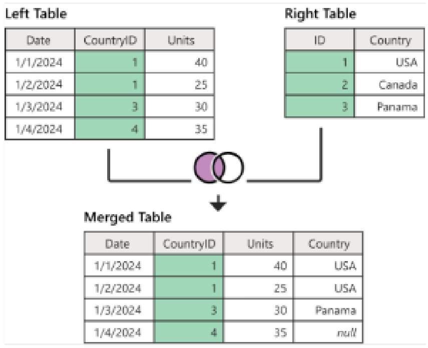
You can use this engine to extract data and transform data.

The Power Query Editor already has a number of in-built transformations.

Power Query is available in products that include Power BI, Excel, Azure Data Factory etc.

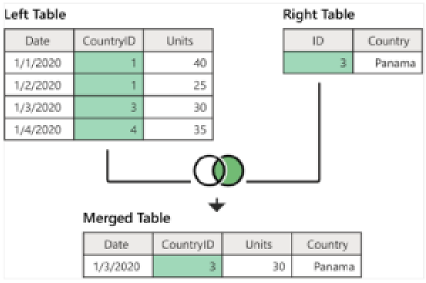
Merge Queries - Here we can join two existing tables based on matching values in one or multiple columns.

Left outer join - This takes all rows from the left table and brings in the matching rows from the right table.



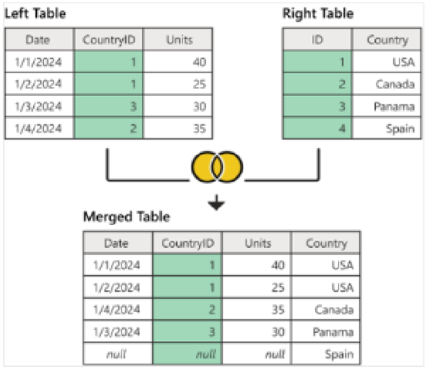
Reference - https://learn.microsoft.com/en-us/power-query/merge-queries-left-outer

Right outer join - This takes all rows from the right table and brings in the matching rows from the left table.



Reference - https://learn.microsoft.com/en-us/power-query/merge-queries-right-outer

<u>Full outer join</u> - This takes all rows from both the left and right table.



Left Table

Date	CountryID	Units
1/1/2024	1	40
1/2/2024	1	25
1/3/2024	3	30
1/4/2024	2	35

Right Table

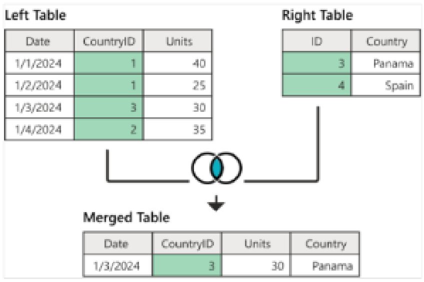
ID	Country
1	USA
2	Canada
3	Panama
4	Spain

Merged Table

Date	CountryID	Units	Country
1/1/2024	1	40	USA
1/2/2024	1	25	USA
1/4/2024	2	35	Canada
1/3/2024	3	30	Panama
null	*null*	*null*	Spain

Reference - https://learn.microsoft.com/en-us/power-query/merge-queries-full-outer

<u>Inner join</u> - This brings only the matching rows from the left and right table.



Left Table

Date	CountryID	Units
1/1/2024	1	40
1/2/2024	1	25
1/3/2024	3	30
1/4/2024	2	35

Right Table

ID	Country
3	Panama
4	Spain

Merged Table

Date	CountryID	Units	Country
1/3/2024	3	30	Panama

Reference - https://learn.microsoft.com/en-us/power-query/merge-queries-inner

Prepare Data - Microsoft Fabric - Data Ingestion and Warehouse

What is Microsoft Fabric

Microsoft Fabric

This is meant to be an end-to-end analytics platform. Here you can perform data movement , data processing, ingesting data, data transformation and report building.

It supports various capabilities that include Data Engineering, Data Science and Real-Time Intelligence.

Reference - https://learn.microsoft.com/en-us/fabric/fundamentals/microsoft-fabric-overview

Different components in Microsoft Fabric

1. **Power BI - Here you can also connect to data sources and visualize your data.**
2. **Databases - You can easily host and manage operational databases.**
3. **You get access to Azure Data Factory - This allows you to ingest, transform and load data.**
4. **You can also ingest data in real time and extract insights and visualizations on the data.**
5. **You can also host a data warehouse. The required compute and storage is managed for you.**
6. **You can also build, deploy and manage your Machine Learning models.**

OneLake

This is one of the core aspects of Microsoft Fabric.

This was meant to provide one unified layer for storing data. Instead of having your data across diverse systems, you can host all of the your data in OneLake.

The data is stored in one format which is the delta parquet format.

OneLake is build on top of Azure Data Lake Storage.

So different types of workloads that use different types of compute engines can all make use of OneLake for storage of data.

Review of the data warehousing fundamentals

Data warehousing fundamentals

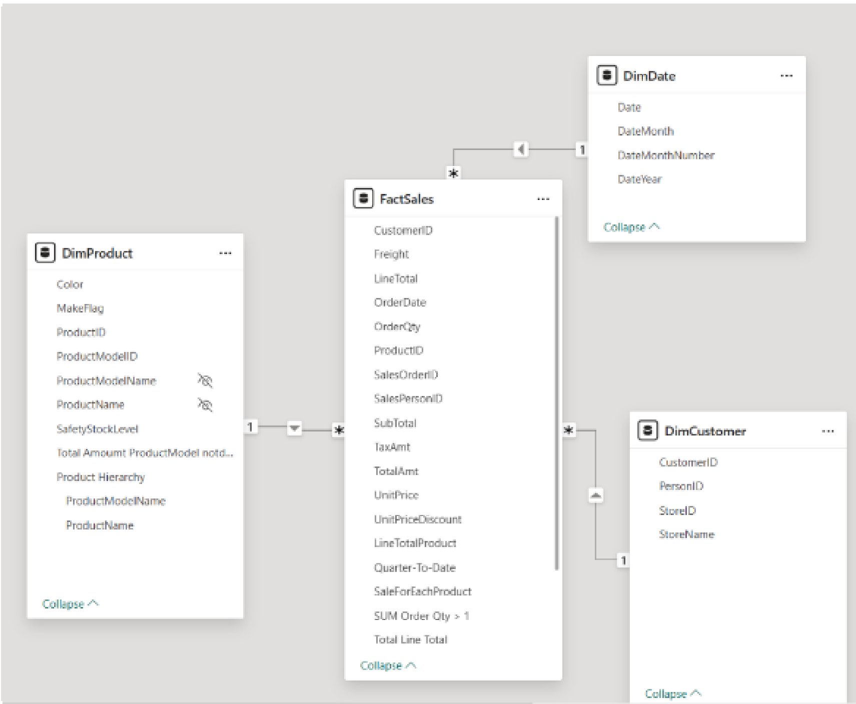
When we design our data model, we seperate our table into fact and dimension tables.

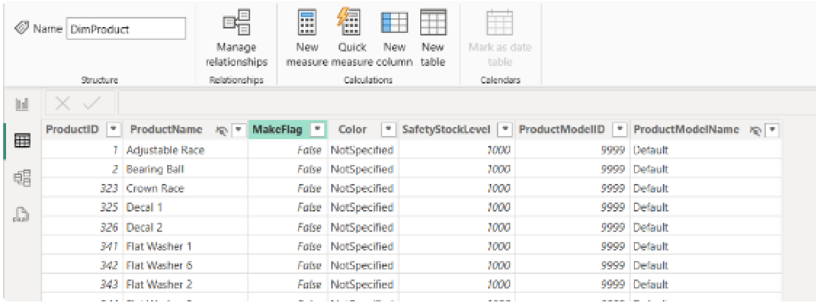
The entire purpose of a data warehouse is we should be able to query and perform analysis over a large data set.

Ou Fact tables contain the actual facts or events. This is normally numerical data that we want to analyze.

Our Dimension tables are used to give a different perspective to our fact-based data.

We can organize our data model in the form of a star schema.





Name: DimProduct | Manage relationships | New measure | Quick measure | New column | New table | Mark as date table

Structure | Relationships | Calculations | Calendars

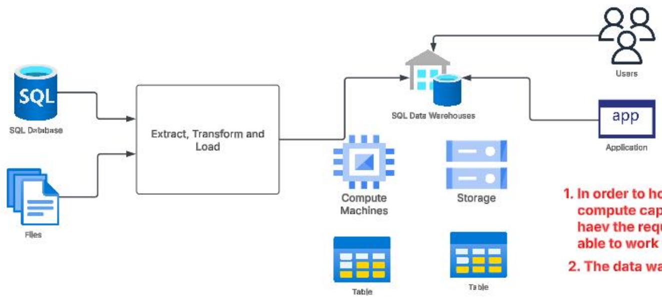
ProductID	ProductName	MakeFlag	Color	SafetyStockLevel	ProductModelID	ProductModelName
1	Adjustable Race	False	NotSpecified	1000	9999	Default
2	Bearing Ball	False	NotSpecified	1000	9999	Default
323	Crown Race	False	NotSpecified	1000	9999	Default
325	Decal 1	False	NotSpecified	1000	9999	Default
326	Decal 2	False	NotSpecified	1000	9999	Default
341	Flat Washer 1	False	NotSpecified	1000	9999	Default
342	Flat Washer 6	False	NotSpecified	1000	9999	Default
343	Flat Washer 2	False	NotSpecified	1000	9999	Default

When it comes to dimension tables, we ideally need to have 2 important columns in addition to other columns that can be used to describe our facts.

Alternate Key - This is sometimes also referred to as the business key. This was the key generated from the source system such as your OLTP system. Here we can liken ProductID as the business key.

Surrogate Key- This can be an automatically generated integer value that helps to uniquely identify each row in the dimension table.

Traditional data warehouse



4. Normally the data warehouse will expose a SQL endpoint. Users and applications can query for the data via the SQL endpoint.

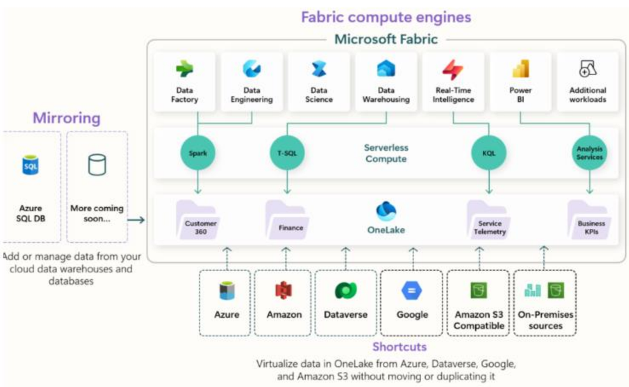
1. In order to host a data warehouse, we need to have compute capacity in the form of machines. This would haev the required database engine capabilities to be able to work with the data.

2. The data warehouse also needs storage for storing the data.

3. Logically your data is stored in the form of fact and dimension tables in the data warehouse.

4. You can have data coming in from different sources. The data can be transformed and then loaded into the data warehouse.

The data warehouse in Microsoft Fabric

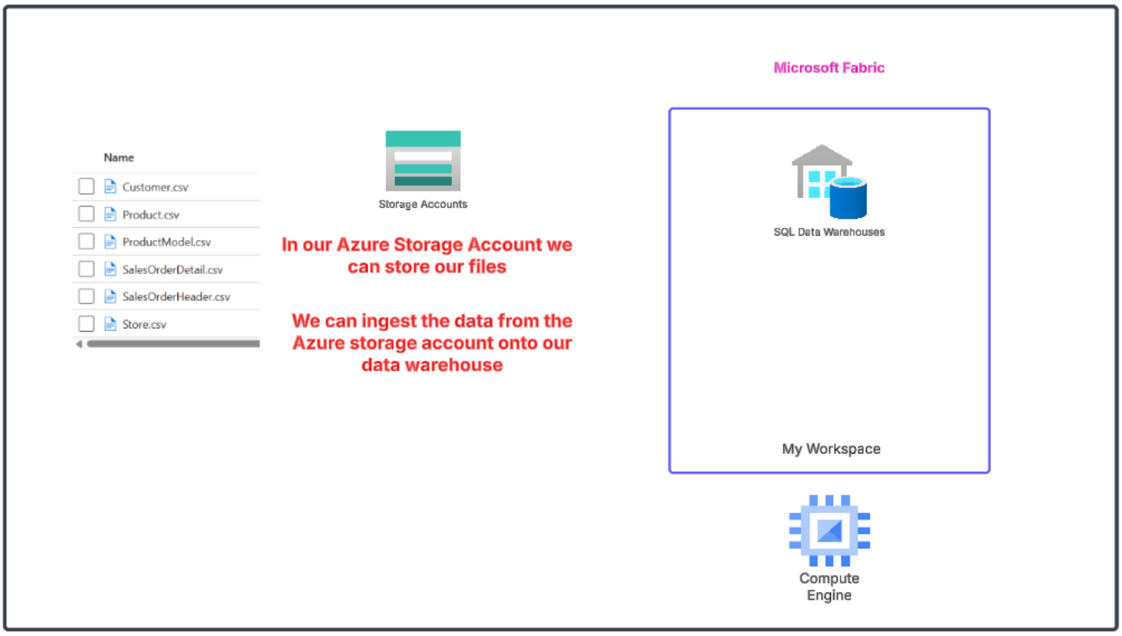


Here we have serverless compute that manages the compute for different workloads. So now this is not only meant for data warehousing , but for other capabilties as well.

Then the storage of data is done on OneLake. This is a central store. SO instead of having one store for your data warehouse, one store for data that is traversed by Notebooks, everything is located in one place.

Reference - https://learn.microsoft.com/en-us/fabric/fundamentals/microsoft-fabric-overview

Microsoft Fabric - Ingesting data



<u>There are different ways in ingesting data</u>

COPY command from T-SQL - You can use this command to copy data from an Azure storage account. This is a high-throughput way in ingesting data.

Data pipelines and Dataflows- Here you get a code-free or a low-code experience in terms of ingesting data.

Cross-warehouse ingestion - You can ingest data from other warehouses as well.

Data warehouse - Slowly Changing Dimensions

Slowly Changing Dimensions

This revolves around the concept on what should we do when certain aspects changes when it comes to Dimension tables.

```
SELECT * FROM DimProduct
```

Messages Results Copy Search

	ProductID	ProductName	MakeFlag	Color	SafetyStockLevel	ProductModelID	ProductModelName
1	722	LL Road Frame - Black, 58	1	Black	500	9	LL Road Frame
2	723	LL Road Frame - Black, 60	1	Black	500	9	LL Road Frame
3	724	LL Road Frame - Black, 62	1	Black	500	9	LL Road Frame
4	736	LL Road Frame - Black, 44	1	Black	500	9	LL Road Frame
5	737	LL Road Frame - Black, 48	1	Black	500	9	LL Road Frame
6	738	LL Road Frame - Black, 52	1	Black	500	9	LL Road Frame

When it comes to Fact tables, these contain actual facts. Here facts are only added over time , but they will not change.

But the data from the source system from where the Dimension tables get populated, those values could change over time.

For example, let's say that for ProductID = 722 , the ProductName just changes slightly over time.

Depending on the type of reports and analysis we conduct, we might need to cater to these changes. Sometime we would need a reference to both the old and new dimension values.

Type 1

Here the old value is completely overwritten by the new value. If the changes does not make a significant difference , we can opt for this.

ProductID	ProductName	Color
722	LL Road Frame - Black, 58A	Black
723	LL Road Frame - Black, 60	Black

Change made to the name of the Product name where the ID is 722.

Type 2

Here we add a new row to the table. Here we want to main the previous version of the data value. Here we need to track a complete history of the changes.

ProductID	ProductName	Color	StartDate	EndDate
722	LL Road Frame - Black, 58	Black	2019-01-01	2022-12-01
722	LL Road Frame - Black, 58A	Black	2023-01-01	NULL

Here we add additional columns to the table to signify the start and end date of the data values.

Type 3

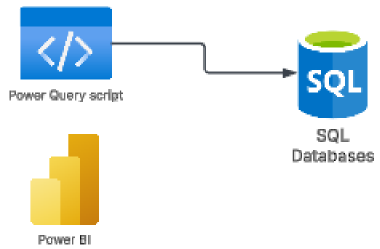
Here we use an additional column to track just the most recent change. We track the current and previous state only.

ProductID	ProductName	Color	PreviousProductName
722	LL Road Frame - Black, 58A	Black	LL Road Frame - Black, 58
723	LL Road Frame - Black, 60	Black	NULL

Microsoft Fabric - Power Query - Query folding

Power Query - Query folding

Power Query is used to perform the required transformations on your data.



If you have a data source of a SQL database, Power Query would send a request to the database to get the data and then the Power Query engine would perform the required transformation.

With Query folding , the Power Query engine will see if there are any steps in the transformation process that could be done by the source database engine.

This could help to optimize the query execution.

Remember that if you are fetching CSV files, basically non-structured data that don't rely on any compute engine, Power Query can't perform query folding.

Indicator	Icon	Description
Folding		The folding indicator tells you that the query up to this step is evaluated by the data source.
Not folding		The not-folding indicator tells you that some part of the query up to this step is evaluated outside the data source. You can compare it with the last folding indicator, if there is one, to see if you can rearrange your query to be more performant.
Might fold		Might fold indicators are uncommon. They mean that a query "might" fold. They indicate either that folding or not folding is determined at runtime, when pulling results from the query, and that the query plan is dynamic. These indicators likely only appear with ODBC or OData connections.
Opaque		Opaque indicators tell you that the resulting query plan is inconclusive for some reason. It generally indicates that there's a true "constant" table, or that that transform or connector isn't supported by the indicators and query plan tool.
Unknown		Unknown indicators represent an absence of a query plan, either due to an error or attempting to run the query plan evaluation on something other than a table (such as a record, list, or primitive).

Reference - https://learn.microsoft.com/en-us/power-query/step-folding-indicators

Prepare Data - Microsoft Fabric – Lakehouse

What is a Lakehouse

Microsoft Fabric Lakehouse

This is a platform that can be used for storing, managing and analyzing both your structured and unstructured data.

All of this can be done from a single location.

It has the capability to manage large volumes of data. You can use tools to analyze and process the data.

The Lakhouse makes use of Spark and SQL engines to process data.

It has a schema-on-read format. This means that you don't need to define the schema beforehand.

It has support for ACID - Atomicity, Consistency, Isolation and Durability.

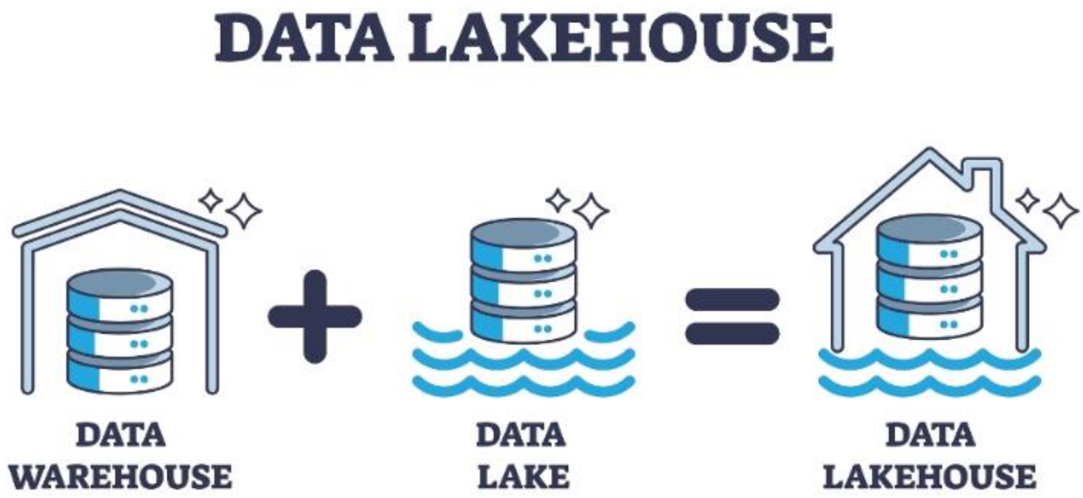
Atomicity - Here each transaction over the data is treated as a single, indivisible unit.

Consistency - Here each transaction will leave the database in a new state that is consistent with defined rules, constraints etc.

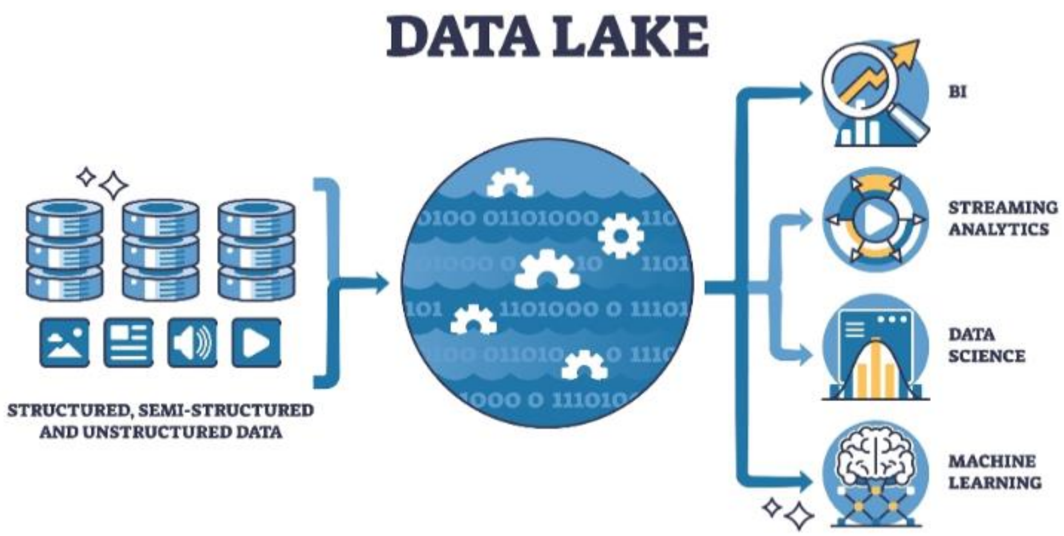
Isolation - Here concurrent transactions can run independently without interfering with each other.

Durability - Once the transaction takes place, it stays committed in the database.

Why do we even consider lakehouses



A data lakehouse is meant to the best of both worlds when it comes to a data lake and a data warehouse.



A data lake is a vast space that is used as a storage layer for your incoming streams of data.

Your data would be in different formats, it would probably be in a raw format. This means you really can't extract any meaningful insights from the data.

There are tools available in the market that can help you to analyze the data in the data lake. That is possible. But normally you might need to shape your data, cleanse it , transform it, shape it for further analysis.

So normally you would use ETL tools to Extract data from a data lake, transform the data and then load the data onto a data warehouse. You can then perform analysis of data within the warehouse.

Problems with a data lake - Governance of data because you just a lot of incoming streams of data from different data sources.

Problems with a data warehouse - It's not cheap. Its expensive to host a data warehouse. You need to make sure the warehouse is updated with data for analysis.

That's why we now have this concept of a lakehouse which is available from different vendors. Here you can store your vast expanse of data. But at the same time, you can model your data there itself , and then perform analysis of your data.

The Lakehouse gives you a SQL analytics endpoint that allows you to query the data. But the tables need to be in a format known as the Delta format.

Normal CSV or Parquet files cannot be queried via the SQL endpoint.

When you drop a file onto the Lakehouse, if the format is valid, the Lakehouse will recognise the file and register it with the metastore.

You can load your data and see your data in the Lakehouse via the Lakehouse explorer.

You can use notebooks to write code that can work with data. This is normally managed by Data Engineers.

You can use data pipelines and Dataflows Gen2 to ingest your data into the lakehouse.

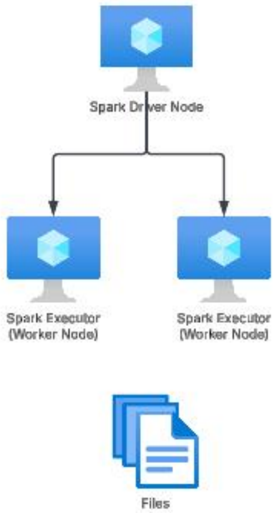
About using Apache Spark on Microsoft Fabric

Apache Spark

This is a distributed computing system that has the capability to process data at scale.

You can feed in batch data, real-time streaming data.

The engine was initially developed at the University of California, but from 2013 the codebase is being maintained by the Apache Software Foundation.



You can install Spark on a cluster of nodes. You have the driver node that is responsible for managing and distributing the tasks across the worker nodes.

The tasks of processing and transforming the data is carried out by the Worker nodes.

Here each worker nodes works in parallel on your data set. Its like a divide and conquer approach which it faster to process large sets of data.

It also supports both in-memory computation and disk-based processing when it comes to large data sets.

There is also support for different programming languages for developing code that can work on the data - Scala, Java, Python, R etc.

Microsoft Fabric also allows you to run code on Spark.

Microsoft Fabric provides a starter pool in each workspace.

You do have the option to customize the settings when it comes to the Spark pool.

You can decide on the type of virtual machines that will be used in the Spark cluster when it comes to the nodes.

You can decide to also autoscale your infrastructure.

The Spark ecosystem on Microsoft Fabric comes with versions of Apache Spark, Delta , Python etc.

You also have a set of libraries that you can make use of.

In otder to interact with Spark, you start writing code in an artefact known as a notebook.

A notebook provides a simple interface for developing code. And using that code which could be in different supported languages, you can start interacting with your data.

Microsoft Fabric - What is an eventhouse

Microsoft Fabric - What is an eventhouse

Eventhouse

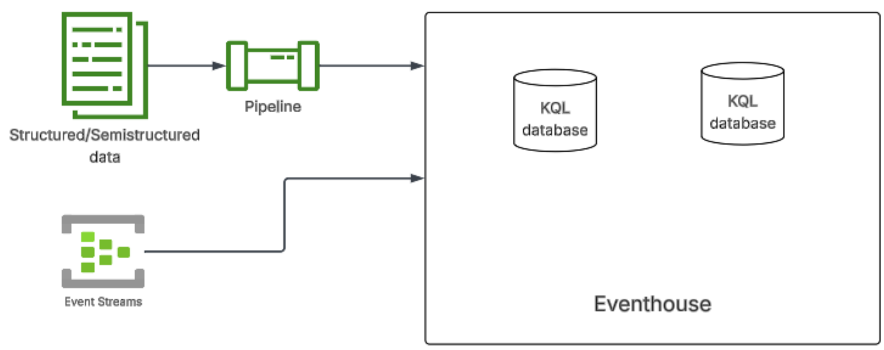
This solution provides an area that allows you to ingest and analyze large volumes of data.

This is ideal to use when you want to analyze your data in real-time. It's ideal for time-series based data.

Within the eventhouse, you can ingest your data into databases.

You can stream in your structured,semistructured and unstructured data.

The compute capacity can be shared across your databases.



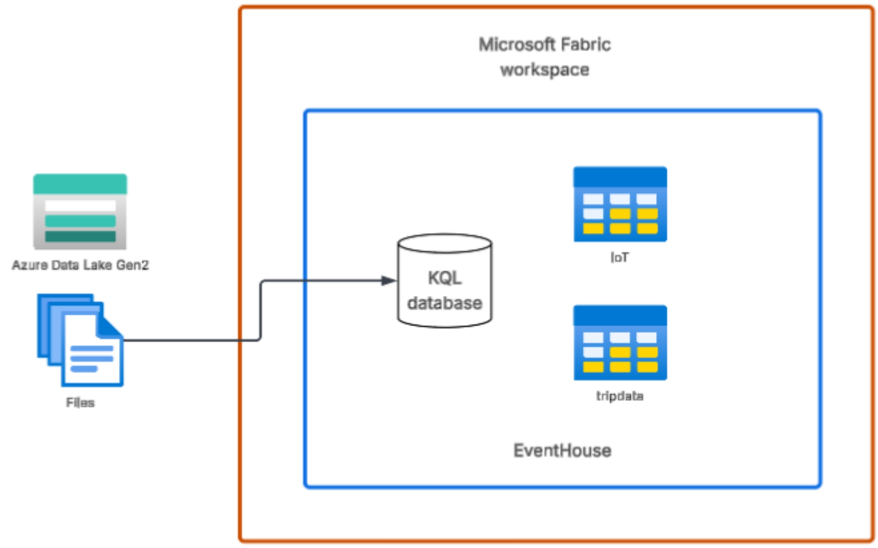
When it comes to allocating resources for the eventhouse, you can set the minimum comsumption level. You can set it to on-demand or onto pre-defined consumption levels. Each level provides capacity units and storage.

Based on the capacity units setup for Microsoft Fabric, you would be entitled for a set of capacity units.

Name	Minimum CUs	SSD capacity (GB) of free storage
Extra extra extra small	2.25	25
Extra extra small	4.25	50
Extra small	8.5	200
Small	13	800
Medium	18	3500-4000
Large	26	5250-6000
Extra large	34	7000-8000
Extra extra large	50	10500-12000

Reference -
https://learn.microsoft.com/en-us/fabric/real-time-intelligence/eventhouse

Lab - Microsoft Fabric - Eventhouse - Ingest data from Azure data lake



We'll setup some parquet-based files in an Azure Data Lake Gen2 storage account.

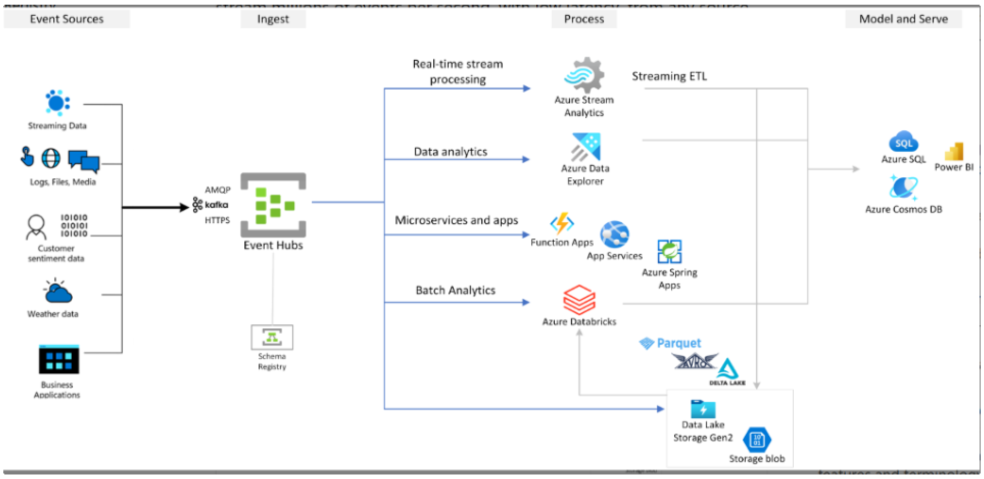
We will ingest the data in the files in a table in the eventhouse in a KQL database.

Microsoft Fabric – Eventstream

Azure Event Hubs

This is a data streaming service that can stream millions of events per second.

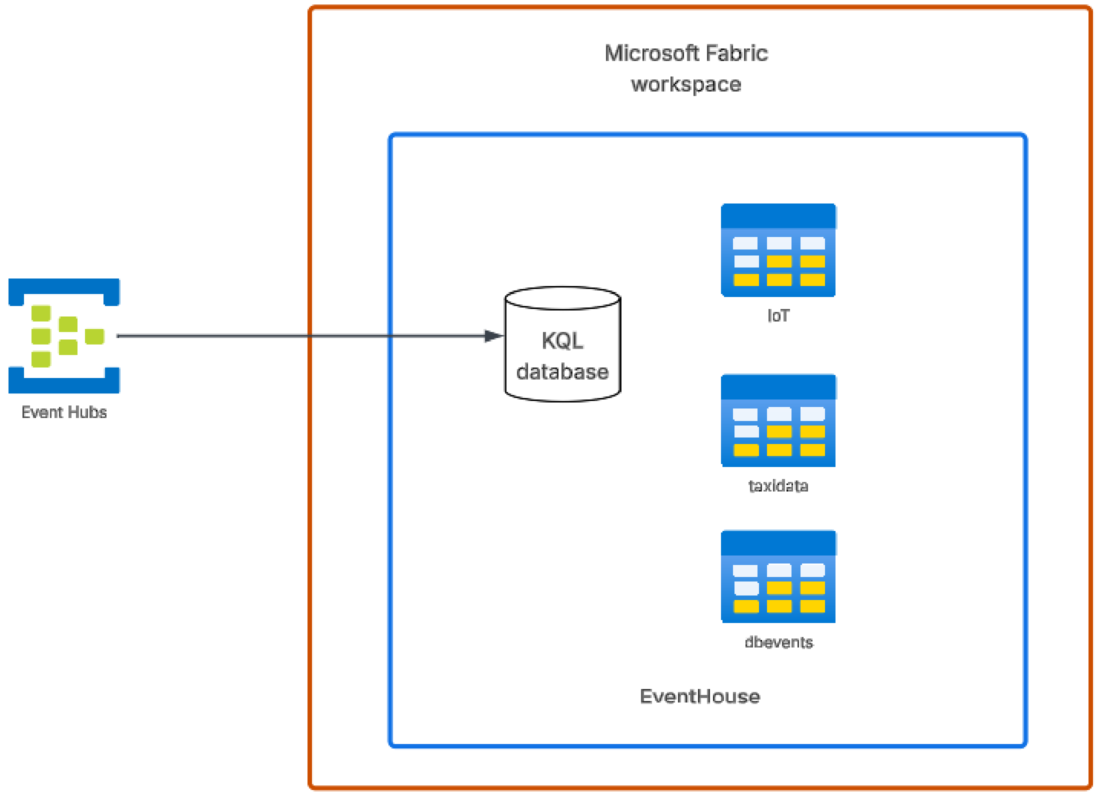
This can be from any source or destination.



Reference - https://learn.microsoft.com/en-us/azure/event-hubs/event-hubs-about

Fabric Eventstream - This allows you to ingest real-time events into Fabric , you can transform them and write them onto a destination. Here you don't need to have prior coding experience to accomplish this.

You have support to get data from a variety of sources - Azure Event Hubs, Azure Event Grid etc.



We'll create an Azure Event Hub resource.

We can generate sample events from Azure Event Hub.

The events can be ingested into the event house.

Maintain a data analytics solution

Medallion Architecture

Medallion Architecture

This is a data design pattern that is used to organize data in a lakehouse.

Here the core goal is to refine your data as it comes in through several layers before it reaches a state where it starts to bring real business value.

You ingest all of your raw data in the bronze layer. The data coming in from different data sources.

This is the landing zone for your data. Here you can have structured, semi-structured or unstructured data.

To ingest data you can use data pipelines, notebooks.

In the Silver area, you have filtered and cleaned data.

Here you will combine and merge data.

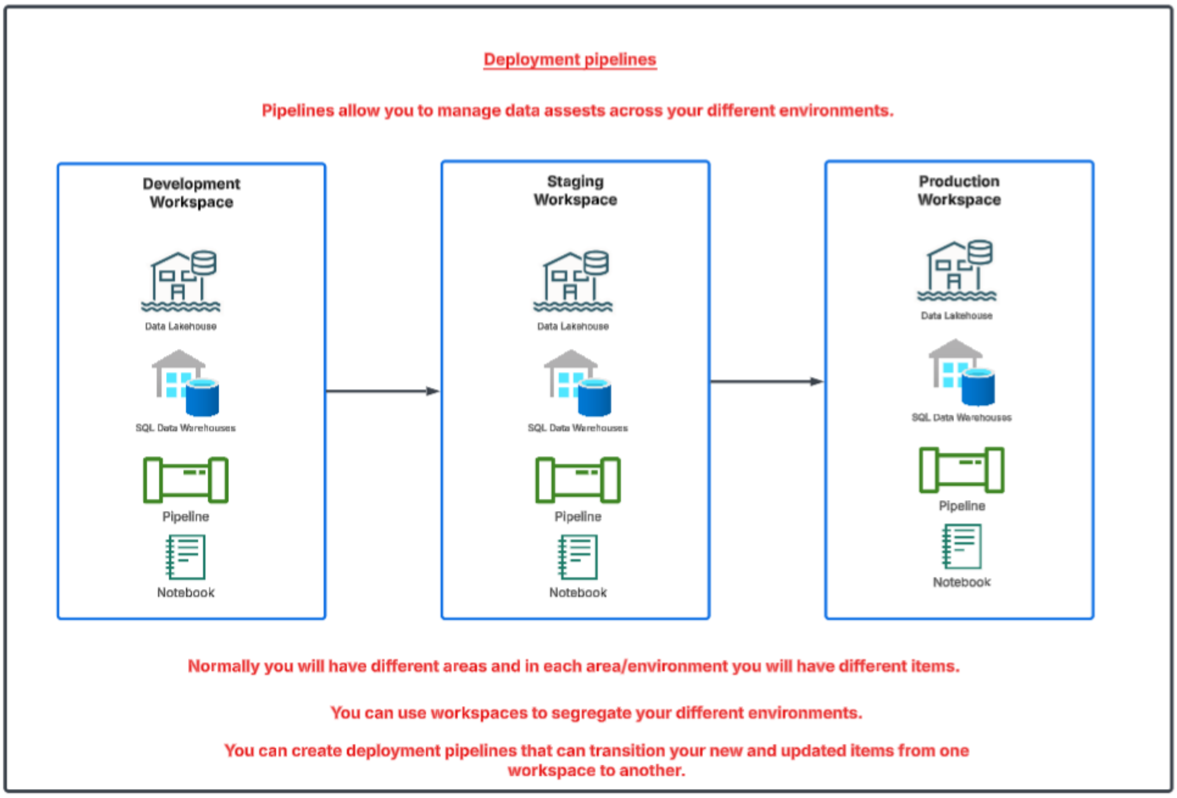
To transform data you can use Data Flow Gen2 , notebooks.

Here you have your curated data that is used to derive business value.

Here refinements have been carried out on the underlying data.

Your gold layer can be in a data warehouse or a lakehouse.

Microsoft Fabric - Deployment pipelines



Supported items

- Activator
- Dashboard
- Data pipeline *(preview)*
- Dataflows gen2 *(preview)*
- Datamart *(preview)*
- Environment *(preview)*
- Eventhouse and KQL database
- EventStream *(preview)*
- KQL Queryset
- Lakehouse *(preview)*
- Mirrored database *(preview)*
- Notebook
- Org app *(preview)*
- Paginated report
- Power BI Dataflow
- Real-time Dashboard
- Report (based on supported semantic models)
- Semantic model (that originates from a .pbix file and isn't a PUSH dataset)
- SQL database *(preview)*
- Warehouse *(preview)*

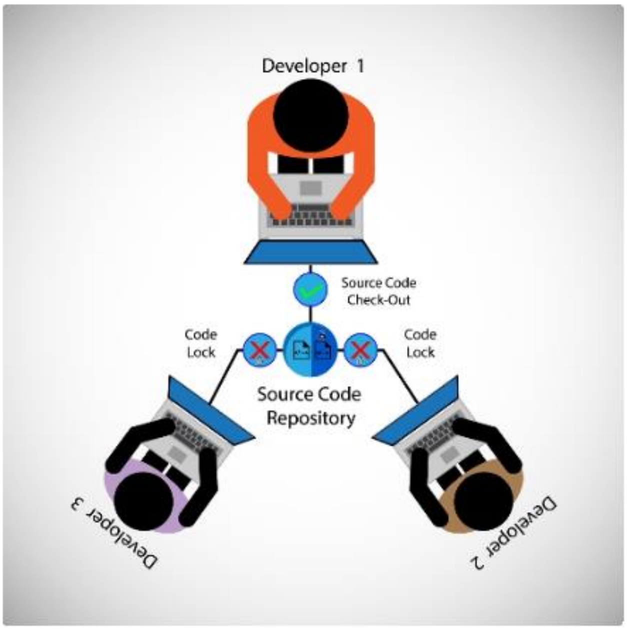
Reference -
https://learn.microsoft.com/en-us/fabric/cicd/deployment-pipelines/intro-to-deployment-pipelines?tabs=new-ui

Microsoft Fabric - Version Control

Version Control

Version control tools are used during the development process.

This allows developers to track changes to different versions of their code base.



Git is a popular version control tool. You can install Git on your local machine and start versioning your code.

You can also have git-based repositories on the Internet via the use of GitHub and Azure Repos.

Microsoft Fabric has support for versioning the items on repositories hosted in GitHub and Azure Repos.

The following items are currently supported:

- Data pipelines *(preview)*
- Dataflows gen2 *(preview)*
- Eventhouse and KQL database *(preview)*
- EventStream *(preview)*
- Lakehouse *(preview)*
- Mirrored database *(preview)*
- Notebooks
- Paginated reports *(preview)*
- Reflex *(preview)*
- Reports (except reports connected to semantic models hosted in Azure Analysis Services, SQL Server Analysis Services, or reports exported by Power BI Desktop that depend on semantic models hosted in MyWorkspace) *(preview)*
- Semantic models (except push datasets, live connections to Analysis Services, model v1) *(preview)*
- Spark Job Definitions *(preview)*
- Spark environment *(preview)*
- SQL database *(preview)*
- Warehouses *(preview)*

Reference -
https://learn.microsoft.com/en-us/fabric/cicd/git-integration/intro-to-git-integration?tabs=azure-devops

Fabric prerequisites

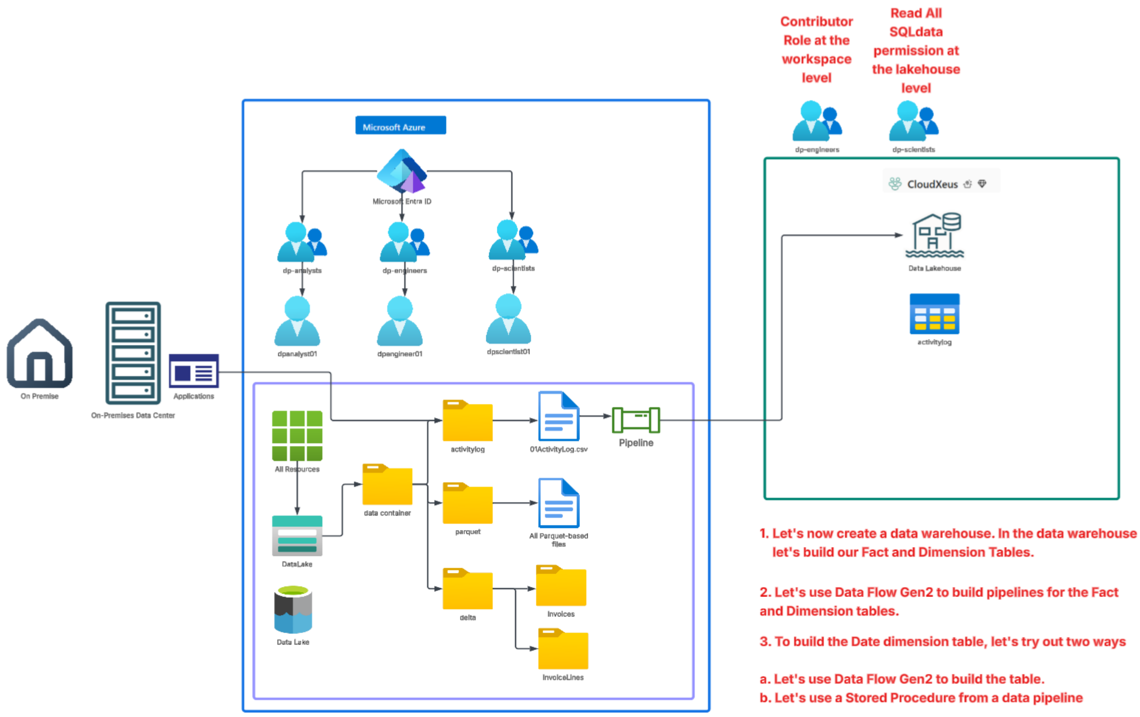
You need to have Fabric capacity. If you have Power BI Premium capacity, that can be used.

The following settings must be enabled on the tenant basis

1. Users can create Fabric items
2.Users can syncrhonize workspace items with their Git repositories

Let's work on a Project

What have we done so far



1. Let's now create a data warehouse. In the data warehouse let's build our Fact and Dimension Tables.

2. Let's use Data Flow Gen2 to build pipelines for the Fact and Dimension tables.

3. To build the Date dimension table, let's try out two ways

a. Let's use Data Flow Gen2 to build the table.
b. Let's use a Stored Procedure from a data pipeline

4. Then let's consolidate all activities in a data pipeline.

5. Then let's see how to work with semantic models attached to the data warehouse.